Logarithms and Their Properties

ANSWER KEY

Section 1: Definition of Logarithm and Inverses

1 Convert $\log_2(32) = 5$ to its equivalent exponential form.

a. $2^5 = 32$ ✓

b. $5^2 = 32$

c. $32^{1/5} = 2$

d. $2 \times 5 = 10$

By definition, $\log_b(y) = x$ is equivalent to $b^x = y$. Here $2^5 = 32$.

2 Evaluate $\log_3\left(\frac{1}{27}\right)$.

a. -3 ✓

b. 3

c. $1/3$

d. $-1/3$

$3^{-3} = \frac{1}{3^3} = \frac{1}{27}$, so $\log_3(1/27) = -3$.

3 What is the inverse function of $f(x) = e^{2x-1}$?

a. $f^{-1}(x) = \frac{\ln x + 1}{2}$ ✓

b. $f^{-1}(x) = \ln(2x - 1)$

c. $f^{-1}(x) = 2\ln x - 1$

d. $f^{-1}(x) = e^{-(2x-1)}$

$y = e^{2x-1} \Rightarrow \ln y = 2x - 1 \Rightarrow 2x = \ln y + 1 \Rightarrow x = \frac{\ln y + 1}{2}$.

Section 2: Product, Quotient, and Power Rules

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

4 Expand $\log_b\left(\frac{x^3y}{z^2}\right)$ into individual logarithms.

a. $3\log_b x + \log_b y - 2\log_b z$ ✓

b. $3\log_b x \cdot \log_b y - 2\log_b z$

c. $6(\log_b x + \log_b y - \log_b z)$

d. $\frac{3\log_b x + \log_b y}{2\log_b z}$

Apply the quotient rule, product rule, and power rule: $\log_b(x^3y) - \log_b(z^2) = 3\log_b x + \log_b y - 2\log_b z$.

5 Condense into a single logarithm: $2\ln(x) + \frac{1}{2}\ln(y) - 3\ln(z)$.

a. $\ln\left(\frac{x^2\sqrt{y}}{z^3}\right)$ ✓

b. $\ln(x^2 + \sqrt{y} - z^3)$

c. $\ln\left(\frac{2x \cdot \frac{1}{2}y}{3z}\right)$

d. $\ln(x^2y^{1/2}z^3)$

Power rule: $\ln(x^2) + \ln(\sqrt{y}) - \ln(z^3) = \ln\left(\frac{x^2\sqrt{y}}{z^3}\right)$.

6 Given $\log_a 2 \approx 0.301$ and $\log_a 3 \approx 0.477$, approximate $\log_a 18$.

a. 1.255 ✓

b. 0.778

c. 1.079

d. 0.954

$\log_a(18) = \log_a(2 \cdot 3^2) = \log_a 2 + 2\log_a 3 \approx 0.301 + 2(0.477) = 0.301 + 0.954 = 1.255$.

Section 3: Change of Base Formula

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

7 State the change of base formula to express $\log_b a$ in terms of natural logarithms.

a. $\log_b a = \frac{\ln a}{\ln b}$ ✓

b. $\log_b a = \frac{\ln b}{\ln a}$

c. $\log_b a = \ln(a - b)$

d. $\log_b a = \ln(a) \cdot \ln(b)$

The change of base formula is $\log_b a = \frac{\log_c a}{\log_c b} = \frac{\ln a}{\ln b}$.

8 Evaluate $\log_4(32)$ using the change of base formula with base 2.

a. $5/2 = 2.5$ ✓

b. 8

c. 2

d. 5/4

$\frac{\log_2 32}{\log_2 4} = \frac{5}{2} = 2.5$.

9 Evaluate the product $\log_2 3 \cdot \log_3 4 \cdot \log_4 8$.

a. 3 ✓

b. 2

c. 4

d. $\log_2 8$ is 3

Using natural logs: $\frac{\ln 3}{\ln 2}\frac{\ln 4}{\ln 3}\frac{\ln 8}{\ln 4} = \frac{\ln 8}{\ln 2} = \log_2 8 = 3$.

Section 4: Domains and Graphs of Logarithmic Functions

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

10 Find the domain and vertical asymptote of $f(x) = \ln(2x - 6)$.

a. **Domain:** $(3, \infty)$**, Asymptote:** $x = 3$ ✓

b. Domain: $[3, \infty)$, Asymptote: $x = 3$

c. Domain: $(6, \infty)$, Asymptote: $x = 6$

d. Domain: $(0, \infty)$, Asymptote: $x = 0$

Argument must be strictly positive: $2x - 6 > 0 \Rightarrow 2x > 6 \Rightarrow x > 3$. Vertical asymptote at $x = 3$.

11 What is the range of any basic logarithmic function $f(x) = \log_b(x)$ $(b > 1)$?

a. $(-\infty, \infty)$ ✓

b. $(0, \infty)$

c. $[0, \infty)$

d. $(-\infty, 0)$

As $x \to 0^+$, $\log_b x \to -\infty$; as $x \to \infty$, $\log_b x \to \infty$. The range is all real numbers.

12 Find the x-intercept of $y = \log_3(x + 2) - 1$.

a. $(1, 0)$ ✓

b. $(0, 0)$

c. $(3, 0)$

d. $(-1, 0)$

Set $y = 0$: $\log_3(x + 2) = 1 \Rightarrow x + 2 = 3^1 = 3 \Rightarrow x = 1$.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647